RACE
Regular Analysis Through Continuous Exercise

Waves on String and Sound Waves : RACE-01

1. A uniform string of length $20m$ is suspended from a rigid support. A short wave pulse is introduced at its lowest end. It starts moving up the string. The time taken to reach the support is :-
(Take $g = 10\ ms^{-2}$)
(A) $\sqrt{2}\,s$ (B) $2\pi\sqrt{2}\,s$ (C) $2s$ (D) $2\sqrt{2}\,s$

PWV289

2. Two identical strings X and Z made of same material have tension T_X and T_Z in them. If their fundamental frequencies are 450 Hz and 300 Hz, respectively, then the ratio T_X/T_Z is :
(A) 0.44 (B) 1.5 (C) 2.25 (D) 1.25

PWV290

3. A uniform thin rope of length 12 m and mass 6 kg hangs vertically from a rigid support and a block of mass 2 kg is attached to its free end. A transverse short wavetrain of wavelength 6 cm is produced at the lower end of the rope. What is the wavelength of the wavetrain (in cm) when it reaches the top of the rope?
(A) 9 (B) 12 (C) 6 (D) 3

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PWV291

4. For a transverse wave travelling along a straight line, the distance between two peaks (crests) is 5 m, while the distance between one crest and one trough is 1.5 m. The possible wavelengths (in m) of the waves are :
(A) 1, 2, 3, ... (B) $\frac{1}{2},\frac{1}{4},\frac{1}{6},\ldots$ (C) 1, 3, 5, ... (D) $\frac{1}{1},\frac{1}{3},\frac{1}{5},\ldots$

PWV292

5. The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4%, will be _____%.

PWV293

6. The mass per unit length of a uniform wire is 0.135 g/cm. A transverse wave of the form $y = -0.21 \sin(x + 30t)$ is produced in it, where x is in meter and t is in second. Then, the expected value of tension in the wire is $x \times 10^{-2}\ N$. Value of x is. (Round-off to the nearest integer)

PWV294

7. The amplitude of wave disturbance propagating in the positive x-direction is given by $y=\frac{1}{(1+x)^2}$ at time $t = 0$ and $y=\frac{1}{1+(x-2)^2}$ at $t = 1s$, where x and y are in metres. The shape of wave does not change during the propagation. The velocity of the wave will be _____m/s.

PWV295

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8. The speed of a transverse wave passing through a string of length 50 cm and mass 10 g is 60 ms^{-1}. The area of cross-section of the wire is 2.0 mm^2 and its Young's modulus is 1.2×10^{11} Nm^{-2}. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}$ m. The value of x is_____.

PWV296

9. The distance between two consecutive points with phase difference of 60° in a wave of frequency 500 Hz is 6.0 m. The velocity with which wave is traveling is _____ km/s.

PWV297

10. A transverse wave is represented by :

$$y = \frac{10}{\pi}\sin\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x\right)$$

For what value of the wavelength the wave velocity is twice the maximum particle velocity?

(A) 40 cm (B) 10 cm (C) 60 cm (D) 20 cm

PWV298

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	B	D	2	1215	2	15	18	A

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RACE
Regular Analysis Through Continuous Exercise

Waves on String and Sound Waves : RACE-02

1. A travelling wave represented by $y = A\sin(\omega t - kx)$ is superimposed on another wave represented by $y = A\sin(\omega t + kx)$. The resultant is :-

(A) A standing wave having nodes at $x = \left(n + \frac{1}{2}\right)\frac{\lambda}{2}, n = 0,1,2$.

(B) A wave travelling along $+x$ direction.

(C) A wave travelling along $-x$ direction.

(D) A standing wave having nodes at $x = \frac{n\lambda}{2}$; n = 0, 1, 2.

PWV299

2. A wire of density $9 \times 10^{-3}\ kg\ cm^{-3}$ is stretched between two clamps 1 m apart. The resulting strain in the wire is 4.9×10^{-4}. The lowest frequency of the transverse vibrations in the wire is (Young's modulus of wire $Y = 9 \times 10^{10}\ Nm^{-2}$), (To the nearest integer),_____.

PWV300

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3. A wire having a linear mass density $9.0 \times 10^{-4}\ kg/m$ is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is _____m.

PWV301

4. The equations of two waves are given by :

$$y_1 = 5\sin 2\pi(x - vt)cm$$

$$y_2 = 3\sin 2\pi(x - vt + 1.5)cm$$

These waves are simultaneously passing through a string. The amplitude of the resulting wave is

(A) 2 cm (B) 4 cm (C) 5.8 cm (D) 8 cm

PWV302

5. In an experiment with sonometer when a mass of 180 g is attached to the string, it vibrates with fundamental frequency of 30 Hz. When a mass m is attached, the string vibrates with fundamental frequency of 50 Hz. The value of m is _____g.

PWV303

6. A guitar string of length 90 cm vibrates with a fundamental frequency of 120 Hz. The length of the string producing a fundamental frequency of 180 Hz will be _____ cm.

PWV304

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7. A wire of length 30 cm, stretched between rigid supports, has it's n^{th} and $(n+1)^{th}$ harmonics at 400 Hz and 450 Hz, respectively. If tension in the string is 2700 N, it's linear mass density is _____ kg/m.

PWV305

8. The fundamental frequency of vibration of a string stretched between two rigid support is 50 Hz. The mass of the string is 18 g and its linear mass density is 20 g/m. The speed of the transverse waves so produced in the string is_____ ms^{-1}.

PWV306

9. The displacement equations of two interfering waves are given by $y_1 = 10 \sin\left(\omega t + \frac{\pi}{3}\right)$ cm, $y_2 = 5[\sin(\omega t) + \sqrt{3}\cos\omega t]cm$ respectively. The amplitude of the resultant wave is _____cm.

PWV307

10. String 1 is connected with string 2. The mass per unit length in string 1 is μ_1 and the mass per unit length in string 2 is 4 μ_1. The tension in the strings is T. A travelling wave is coming from the left. What fraction of the energy in the incident wave goes into string 2?



(A) 1/8 (B) 4/9 (C) 2/3 (D) 8/9

PWV308

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	35	10	A	500	60	3	90	20	D

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Waves on String and Sound Waves : RACE-03

1. The total length of a sonometer wire between fixed ends is 110 *cm*. Two bridges are placed to divide the length of wire in ratio 6 : 3 : 2. The tension in the wire is 400 *N* and the mass per unit length is 0.01 kg/m. What is the minimum common frequency with which three parts can vibrate ?

(A) 166 *Hz* (B) 1000 *Hz* (C) 1100 *Hz* (D) 100 *Hz*

PWV309

2. A sound wave of frequency 245 *Hz* travels with the speed of 300 ms^{-1} along the positive *x*-axis. Each point of the wave moves to and fro through a total distance of 6 *cm*. What will be the mathematical expression of this travelling wave ?

(A) $Y(x, t) = 0.03\ [\sin 5.1\, x - (0.2 \times 10^3)t]$ (B) $Y(x, t) = 0.06\ [\sin 5.1\, x - (1.5 \times 10^3)t]$

(C) $Y(x, t) = 0.06\ [\sin 0.8\, x - (0.5 \times 10^3)t]$ (D) $Y(x, t) = 0.03\ [\sin 5.1\, x - (1.5 \times 10^3)t]$

PWV310

3. A longitudinal wave is represented by $x = 10 \sin 2\pi \left(nt - \frac{x}{\lambda}\right) cm$. The maximum particle velocity will be four times the wave velocity if the determined value of wavelength is equal to :

(A) 2π (B) 5π (C) π (D) $\frac{5\pi}{2}$

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PWV311

4. The ratio of speed of sound in hydrogen gas to the speed of sound in oxygen gas at the same temperature is

(A) 4 : 1 (B) 1 : 2 (C) 1 : 4 (D) 1 : 1

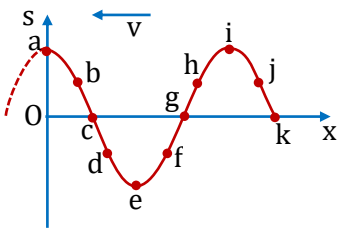
PWV312

5. A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s, the frequency difference between the fundamental and second harmonic of this pipe is _____ *Hz*.

PWV313

Paragraph for Questions No. 6 to 10

The figure represents the instantaneous displacement-position graph of a longitudinal harmonic wave travelling along the negative *x*-axis. Identify the **correct statement(s)** related to the movement of the points shown in the figure.



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6. The points moving in the direction of wave are:
(A) b (B) c (C) f (D) i

PWV314

7. The points moving opposite to the direction of propagation are:
(A) a (B) d (C) f (D) j

PWV315

8. The stationary points are:
(A) a (B) c (C) g (D) k

PWV316

9. The points of maximum compression are:
(A) c (B) g (C) e (D) k

PWV317

10. The points of maximum rarefaction are:
(A) a (B) e (C) g (D) i

PWV318

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	B	A	106.00 to 107.20	AB	C	A	AD	C

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RACE
Regular Analysis Through Continuous Exercise

Waves on String and Sound Waves : RACE-04

1. **Statement-1**: Two longitudinal waves given by equations: $y_1(x, t) = 2a \sin(\omega t - kx)$ and $y_2(x, t) = a \sin(2\omega t - 2kx)$ will have equal intensity.

Statement-2: Intensity of waves of given frequency in same medium is proportional to square of amplitude only.

(A) Statement-1 is false, statement-2 is true.

(B) Statement-1 is true, statement-2 is false.

(C) Statement-1 is true, statement-2 true; statement-2 is the correct explanation of statement-1

(D) Statement-1 is true, statement-2 is true; statement -2 is not correct explanation of statement-1.

PWV319

2. Three harmonic waves having equal frequency ν and same intensity I_0, have phase angles 0, $\frac{\pi}{4}$ and $-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to :

(A) 5.8 I_0 (B) 0.2 I_0 (C) I_0 (D) 3 I_0

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PWV320

3. A student is performing the experiment of resonance column. The diameter of the column tube is 6 *cm*. The frequency of the tuning fork is 504 *Hz*. Speed of the sound at the given temperature is 336 *m/s*. The zero of the meter scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is:

(A) 13 *cm* (B) 16.6 *cm* (C) 18.4 *cm* (D) 14.8 *cm*

PWV321

4. A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open pipe is $\frac{x}{3}L\sqrt{\frac{\rho_1}{\rho_2}}$ where x is _____. (Round off to the Nearest Integer)

PWV322

5. For a certain organ pipe, the first three resonance frequencies are in the ratio of 1 : 3 : 5 respectively. If the frequency of fifth harmonic is 405 *Hz* and the speed of sound in air is 324 ms^{-1} the length of the organ pipe is _____*m*.

PWV323

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6. The end correction of a resonance column is 1 *cm*. If the shortest length resonating with the tuning fork is 10 *cm*, the next resonating length should be :-

(A) 36 *cm* (B) 40 *cm* (C) 28 *cm* (D) 32 *cm*

PWV324

7. A tuning fork is vibrating at 250 *Hz*. The length of the shortest closed organ pipe that will resonate with the tuning fork will be _____ *cm*. (Take speed of sound in air as 340 ms^{-1})

PWV325

8. In an experiment to determine the velocity of sound in air at room temperature using a resonance is observed when the air column has a length of 20.0 *cm* for a tuning fork of frequency 400 *Hz* is used. The velocity of the sound at room temperature is 336 ms^{-1}. The third resonance is observed when the air column has a length of _____*cm*.

PWV326

9. A tuning fork of frequency 340 *Hz* resonates in the fundamental mode with an air column of length 125 *cm* in a cylindrical tube closed at one end. When water is slowly poured in it, the minimum height of water required for observing resonance once again is _____*cm*.

(Velocity of sound in air is 340 ms^{-1})

PWV327

10. The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is 20 *cm*. The length of the open organ pipe is _____ *cm*.

PWV328

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	D	4	1	D	34	104	50	80

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RACE
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Waves on String and Sound Waves : RACE-05

1. The velocity of sound in a gas. in which two wavelengths 4.08 m and 4.16 m produce 40 beats in 12s, will be :

(A) 2.82.8 ms^{-1} (B) 175.5 ms^{-1} (C) 353.6 ms^{-1} (D) 707.2 ms^{-1}

PWV329

2. Two cars are approaching each other at an equal speed of 7.2 km/hr. When they see each other, both blow horns having frequency of 676 Hz. The beat frequency heard by each driver will be ____ Hz. [Velocity of sound in air is 340 m/s.]

PWV330

3. A tuning fork A of unknown frequency produces 5 beats/s with a fork of known frequency 340 Hz. When fork A is filed, the beat frequency decreases to 2 beats/s. What is the frequency of fork A ?

(A) 342 Hz (B) 345 Hz (C) 335 Hz (D) 338 Hz

PWV331

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4. The frequency of a car horn encountered a change from 400 Hz to 500 Hz. When the car approaches a vertical wall. If the speed of sound is 330 m/s. Then the speed of car is _____ km/h.

PWV332

5. When a car is approaching the observer, the frequency of horn is 100 Hz. After passing the observer, it is 50 Hz. If the observer moves with the car, the frequency will be $\frac{x}{3}$ Hz where x = _____.

PWV333

6. An observer is riding on a bicycle and moving towards a hill at 18 kmh^{-1}. He hears a sound from a source at some distance behind him directly as well as after its reflection from the hill. If the original frequency of the sound as emitted by source is 640 Hz and velocity of the sound in air is 320 m/s, the beat frequency between the two sounds heard by observer will be ____ Hz.

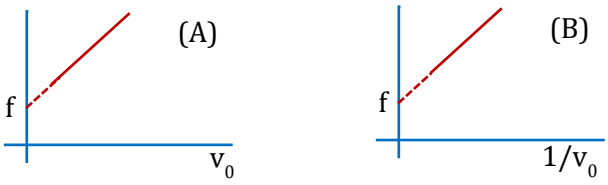
PWV334

7. A set of 20 tuning forks are arranged in a series of increasing frequencies. If each fork gives 4 beats with respect to the preceding fork and the frequency of the last fork is twice the frequency of the first, then the frequency of last fork is____Hz.

PWV335

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8. A source of sound emits sound waves at frequency f_0. It is moving towards an observer with fixed speed v_s($v_s < v$, where v is the speed of sound in air). If the observer were to move towards the source with speed v_0, one of the following two graphs (*A* and *B*) will give the correct variation of the frequency *f* heard by the observer as v_0 is changed.



The variation of *f* with v_0 is given correctly by :-

(A) Graph *A* with slope = $\frac{f_0}{(v+v_s)}$ (B) Graph *B* with slope = $\frac{f_0}{(v-v_s)}$

(C) Graph *B* with slope = $\frac{f_0}{(v+v_s)}$ (D) Graph *A* with slope = $\frac{f_0}{(v-v_s)}$

PWV336

9. Two cars *X* and *Y* are approaching each other with velocities 36 km/h and 72 km/h respectively. The frequency of a whistle sound as emitted by a passenger in car *X*, heard by the passenger in car *Y* is 1320 *Hz*. If the velocity of sound in air is 340 m/s, the actual frequency of the whistle sound produced is _____ *Hz*.

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PWV337

10. For a specific wavelength 670 *nm* of light coming from a galaxy moving with velocity *v*, the observed wavelength is 670.7 *nm*.

The value of *v* is :

(A) $3 \times 10^8\ ms^{-1}$ (B) $3 \times 10^{10}\ ms^{-1}$ (C) $3.13 \times 10^5\ ms^{-1}$ (D) $4.48 \times 10^5\ ms^{-1}$

PWV338

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	8	C	132	200	20	152	D	1210	C

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